

**More Serious Algebra I**

26/27 Semester 1

| <b>Week</b> | <b>Concepts (Not necessarily accurate)</b>                                 |
|-------------|----------------------------------------------------------------------------|
| 3           | Group axioms, subgroup tests, cyclic groups, element order                 |
| 4           | Dihedral group $D_n$ , products, inverses, cyclic subgroups, orders        |
| 5           | Subgroup order, generators, cyclic groups                                  |
| 6           | Left/right cosets, partitions, index, Lagrange's theorem                   |
| 7           | Normality tests, cosets, quotient operations                               |
| 8           | Kernel, image, injectivity/surjectivity, normal subgroups, quotient groups |
| 9           | Everything from Weeks 3–9                                                  |
| 10          | Ring axioms, units, zero divisors, subring tests                           |
| 11          | Ideals, principal ideals in familiar rings, subrings vs. ideals            |
| 12          | Kernels, images, quotient rings                                            |
| 13          | Divisibility, degree, remainders, roots/factors                            |
| 14          | GCD, Bézout identity, polynomial division                                  |

**Format.**

- (1) Four problems each week for two categories: Starter and Advanced.
- (2) Four easy for Starter.
- (3) One low-medium, two medium, and one hard for Advanced.
- (4) The problems will be given on the spot.

**Time and Location.** Tuesday, 2:00-3:40 PM, any empty room.

## Unused Problems

This is the current draft of the list of rejected candidates.

**1 (E).** Let  $G$  be a group. Show that  $(ab)^{-1} = b^{-1}a^{-1}$  for all  $a, b \in G$ .

**2 (L-M).** A group that has only a finite number of subgroups must be finite.

**3 (M).** Let  $G$  be a nonempty finite set with an associative binary operation such that for all  $a, b, c \in G$ ,  $ab = ac \Rightarrow b = c$  and  $ba = ca \Rightarrow b = c$  (in other words,  $G$  satisfies left and right cancellation laws). Then  $G$  is a group. Show that this conclusion may be false if  $G$  is infinite.

**4 (H).** Let the group  $G$  be the union of finitely many, let us say  $n$ , cosets of (not necessarily distinct) subgroups  $C_1, C_2, \dots, C_n$ :

$$G = \bigcup_{i=1}^n C_i g_i.$$

Then at least one subgroup  $C_i$  has finite index in  $G$ .

**5 (L-M).** Assume  $G = \{1, a, b, c\}$  is a group of order 4 with identity 1. Assume also that  $G$  has no elements of order 4. Use the cancellation laws to show that there is a unique Cayley table for  $G$ . Deduce that  $G$  is abelian.

**6 (E).** Let  $G$  be a finite group.

(a) Let  $a \in G$ . Show that there is a positive integer  $n$  such that  $a^n = e$ , the identity element.

(b) Show that there exists a positive integer  $n$  such that  $a^n = e$  for all  $a \in G$ .

**7 (H).** Let  $G$  be the set of integers modulo  $2^n$  with operation

$$x * y = 4xy + x(-1)^y + y(-1)^x \pmod{2^n}.$$

Show that  $G$  is a cyclic group.